

Script Interpreter

./test1.sh

```
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ ./test1.sh
total 244
-rw-r--r-- 1 tunabeluga tunabeluga 538 Oct 14 19:40 Makefile
-rw-r--r-- 1 tunabeluga tunabeluga 647 Oct 16 22:31 app.c
-rw-r--r-- 1 tunabeluga tunabeluga 1992 Oct 16 23:06 app.o
-rw-r--r-- 1 tunabeluga tunabeluga 158669 Oct 14 19:40 ece357ps3.pdf
-rw-r--r-- 1 tunabeluga tunabeluga 850 Oct 16 21:35 ls.out
-rwxr-xr-x 1 tunabeluga tunabeluga 21552 Oct 16 23:06 myshell
-rw-r--r-- 1 tunabeluga tunabeluga 10516 Oct 16 23:16 myshell.c
-rw-r--r-- 1 tunabeluga tunabeluga 958 Oct 16 22:17 myshell.h
-rw-r--r-- 1 tunabeluga tunabeluga 11704 Oct 16 23:06 myshell.o
-rwxr-xr-x 1 tunabeluga tunabeluga 35 Oct 16 22:29 script.sh
-rwxr-xr-x 1 tunabeluga tunabeluga 310 Oct 14 22:50 test1.sh
-rwxr-xr-x 1 tunabeluga tunabeluga 639 Oct 14 22:50 test2.sh
-rw-r--r-- 1 tunabeluga tunabeluga 18 Oct 16 23:35 testfile.out
-rw-r--r-- 1 tunabeluga tunabeluga 18 Oct 16 23:35 testfile2.out
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ echo $?
17
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ cat testfile.out
TESTINGTESTING123
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ cat testfile2.out
TESTINGTESTING123
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$
```

./test2.sh

```
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ ./test2.sh
/tmp
/home/tunabeluga
./test2.sh: line 14: lssssssss: command not found
./test2.sh: line 17: testfile.out: Permission denied
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ echo $?
1
```

Interactive Shell:

```
(base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ ./myshell
myshell: ls
Makefile app.c app.o blah ece357ps3.pdf ls.out myshell myshell.c myshell.h myshell.o script.sh test1.sh test2.sh testfile
.out testfile2.out
Child process 327021 exited normally
Real: 0.001292s User: 0.001179s Sys: 0.000000s
myshell: cd blah
myshell: pwd
/home/tunabeluga/ECE357/ece357/pset03/blah
myshell: cd ..
myshell: pwd
/home/tunabeluga/ECE357/ece357/pset03
myshell: ls -l
total 256
-rw-r--r-- 1 tunabeluga tunabeluga 538 Oct 14 19:40 Makefile
-rw-r--r-- 1 tunabeluga tunabeluga 699 Oct 17 12:34 app.c
-rw-r--r-- 1 tunabeluga tunabeluga 2240 Oct 17 13:59 app.o
drwxr-xr-x 2 tunabeluga tunabeluga 4096 Oct 17 13:59 blah
-rw-r--r-- 1 tunabeluga tunabeluga 158669 Oct 14 19:40 ece357ps3.pdf
-rw-r--r-- 1 tunabeluga tunabeluga 850 Oct 16 21:35 ls.out
-rwxr-xr-x 1 tunabeluga tunabeluga 21832 Oct 17 13:59 myshell
-rw-r--r-- 1 tunabeluga tunabeluga 12675 Oct 17 13:52 myshell.c
-rw-r--r-- 1 tunabeluga tunabeluga 976 Oct 17 11:42 myshell.h
-rw-r--r-- 1 tunabeluga tunabeluga 12752 Oct 17 13:59 myshell.o
-rwxr-xr-x 1 tunabeluga tunabeluga 35 Oct 17 02:05 script.sh
-rwxr-xr-x 1 tunabeluga tunabeluga 310 Oct 14 22:50 test1.sh
-rwxr-xr-x 1 tunabeluga tunabeluga 639 Oct 14 22:50 test2.sh
-rw-r--r-- 1 tunabeluga tunabeluga 18 Oct 17 12:33 testfile.out
-rw-r--r-- 1 tunabeluga tunabeluga 18 Oct 17 12:33 testfile2.out
Child process 327158 exited normally
Real: 0.001978s User: 0.001179s Sys: 0.001847s
```

```
myshell: ls -l >ls.out
Child process 327179 exited normally
Real: 0.001986s User: 0.003033s Sys: 0.001847s
myshell: cat ls.out
total 252
-rw-r--r-- 1 tunabeluga tunabeluga 538 Oct 14 19:40 Makefile
-rw-r--r-- 1 tunabeluga tunabeluga 699 Oct 17 12:34 app.c
-rw-r--r-- 1 tunabeluga tunabeluga 2240 Oct 17 13:59 app.o
drwxr-xr-x 2 tunabeluga tunabeluga 4096 Oct 17 13:59 blah
-rw-r--r-- 1 tunabeluga tunabeluga 158669 Oct 14 19:40 ece357ps3.pdf
-rw-r--r-- 1 tunabeluga tunabeluga 0 Oct 17 14:01 ls.out
-rwxr-xr-x 1 tunabeluga tunabeluga 21832 Oct 17 13:59 myshell
-rw-r--r-- 1 tunabeluga tunabeluga 12675 Oct 17 13:52 myshell.c
-rw-r--r-- 1 tunabeluga tunabeluga 976 Oct 17 11:42 myshell.h
-rw-r--r-- 1 tunabeluga tunabeluga 12752 Oct 17 13:59 myshell.o
-rwxr-xr-x 1 tunabeluga tunabeluga 35 Oct 17 02:05 script.sh
-rwxr-xr-x 1 tunabeluga tunabeluga 310 Oct 14 22:50 test1.sh
-rwxr-xr-x 1 tunabeluga tunabeluga 639 Oct 14 22:50 test2.sh
-rw-r--r-- 1 tunabeluga tunabeluga 18 Oct 17 12:33 testfile.out
-rw-r--r-- 1 tunabeluga tunabeluga 18 Oct 17 12:33 testfile2.out
Child process 327228 exited normally
Real: 0.001302s User: 0.004179s Sys: 0.001847s
myshell: 
```

Invalid ls command along with exit status from main shell

```
myshell: ls jeffhaknerisnotcarlsable
ls: cannot access 'jeffhaknerisnotcarlsable': No such file or directory
Child process 327939 exited with return value 2
Real: 0.001266s User: 0.001146s Sys: 0.000000s
myshell: exit
• (base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ echo $?
2
```

Dumpcore (dereferences NULL pointer -> seg fault) along with exit status from main shell

```
myshell: ./dumpcore
Child process 328471 exited with signal 11 (Segmentation fault)
Real: 0.004169s User: 0.000822s Sys: 0.000000s
myshell: exit
• (base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ echo $?
139
```

****Extra Credit:**

Part (a)

```
myshell: ls ece357ishard
ls: cannot access 'ece357ishard': No such file or directory
Child process 328869 exited with return value 2
Real: 0.001635s User: 0.001456s Sys: 0.000000s
myshell: echo $?
2
```

Part (b) Badda Bing Badda Boom

```
• (base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ cat script.sh
#!/myshell
echo badda $1 badda $2
• (base) tunabeluga@AndrewGaming:~/ECE357/ece357/pset03$ ./script.sh bing boom
badda bing badda boom
Child process 329116 exited normally
Real: 0.001677s User: 0.000932s Sys: 0.000000s
```

Part (c)

```
myshell: export PS1=C:\>
C:\> ls
Makefile  app.o  dumpcore  ece357ps3.pdf  myshell  myshell.h  script.sh  test2.sh  testfile2.out
app.c     blah  dumpcore.c  ls.out        myshell.c  myshell.o  test1.sh  testfile.out
Child process 329623 exited normally
Real: 0.001418s User: 0.001286s Sys: 0.000000s
C:\> export PS1=Peekaboo:
Peekaboo: █
```

Part (d)

```
myshell: echo $HOME
/home/tunabeluga
Child process 329928 exited normally
Real: 0.006003s User: 0.001611s Sys: 0.000000s
myshell: echo $PS1

Child process 329974 exited normally
Real: 0.001260s User: 0.002705s Sys: 0.000000s
myshell: echo $PATH
/home/tunabeluga/.vscode-server/bin/03c
Child process 330026 exited normally
Real: 0.001073s User: 0.003660s Sys: 0.000000s
myshell: █
```

PART (e) NEXT PAGE

Part (e) - using echo and ls along with wildcards

```
myshell: echo *.c
app.c dumpcore.c myshell.c
Child process 330828 exited normally
Real: 0.000932s User: 0.000763s Sys: 0.000000s
myshell: echo *.h
myshell.h
Child process 330829 exited normally
Real: 0.000840s User: 0.001494s Sys: 0.000000s
myshell: echo *.c *.h
app.c dumpcore.c myshell.c myshell.h
Child process 330845 exited normally
Real: 0.001198s User: 0.001494s Sys: 0.001079s
myshell: ls -l *.c *.h
-rw-r--r-- 1 tunabeluga tunabeluga 699 Oct 17 12:34 app.c
-rw-r--r-- 1 tunabeluga tunabeluga 82 Oct 17 14:19 dumpcore.c
-rw-r--r-- 1 tunabeluga tunabeluga 12735 Oct 17 14:23 myshell.c
-rw-r--r-- 1 tunabeluga tunabeluga 976 Oct 17 11:42 myshell.h
Child process 330891 exited normally
Real: 0.001711s User: 0.003112s Sys: 0.001079s
myshell: █
```